F18CF - Assignment 3

Instructions:

- This assignment is due at 23:59 (Edinburgh Time) on the date specified above
- Your solutions can be written or typeset. No paper submission is necessary
- *Gradescope* submission link available on assignment page in Canvas
- The assessment criteria are listed on the assignment page
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 5 questions worth a total of 40 marks and 2 Bonus marks. Marks are listed with each question.
- An additional 1 mark is awarded for meeting the submission standards.

Questions:

11 marks 1. (a) Find the eigenvalues of $A = \begin{pmatrix} 1 & -1 & 0 \\ 4 & -3 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ and determine whether A is diagonalisable

Solution: The Characteristic polynomial of A (expanded along the third column) is give by:

$$\begin{aligned}
\chi_A(\lambda) = \det(A - \lambda I_3) &= \det \begin{pmatrix} 1-\lambda & -1 & 0 \\ 4 & -3-\lambda & 0 \\ 0 & 1 & 1-\lambda \end{pmatrix} \\
&= (1-\lambda) \begin{vmatrix} 1-\lambda & -1 \\ 4 & -3-\lambda \end{vmatrix} \\
&= (1-\lambda)\left[(1-\lambda)(-3-\lambda) - (-1)(4)\right] \\
&= (1-\lambda)\left[-3 + 3\lambda + \lambda + \lambda^2 + 4\right] \\
&= (1-\lambda)\left[\lambda^2 + 2\lambda + 1\right] \\
&= (1-\lambda)(1+\lambda)^2
\end{aligned}$$

which means the eigenvalues are $\lambda = 1, -1$. Since the algebraic multiplicity for $\lambda = -1$ is greater than 1, we have to check the geometric multiplicity too. The augmented matrix associate with the eigenspace corresponding to $\lambda = -1$ is given by:

$$\begin{aligned}
\left(\begin{array}{ccc|c} 1-(-1) & -1 & 0 & 0 \\ 4 & -3-(-1) & 0 & 0 \\ 0 & 1 & 1-(-1) & 0 \end{array}\right) &= \left(\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right) \\
&\sim \left(\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right) \qquad R_2 \to R_2 - 2R_1 \\
&\sim \left(\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \qquad R_2 \leftrightarrow R_3
\end{aligned}$$

The REF has exactly one free variable. So the dimension of E_{-1} (and hence geometric multiplicity) is one, which disagrees with the algebraic multiplicity of $\lambda = -1$. Therefore this matrix is NOT diagonalisable.

(b) Consider the vector space of solutions W to the differential equation

$$y^{(3)} + 7y^{(2)} + 14y' + 8y = 0$$

We can assume without proof that a basis for W is $\mathcal{B} = \left(e^{-x}, e^{-2x}, e^{-4x}\right)$. Define a inner product for $f, g \in W$ as

$$\langle f, g \rangle = \int_0^\infty f(x)g(x)dx$$

i. For a general real constant $k > 0$, evaluate the following integral to show that

$$\int_0^\infty e^{-kx}dx = \frac{1}{k}$$

Solution:

$$\begin{aligned}
\int_0^\infty e^{-kx}dx &= \lim_{b\to\infty}\left[-\frac{1}{k}e^{-kx}\right]_0^b \\
&= \lim_{b\to\infty}\left[-\frac{1}{k}e^{-kb}\right] - \left[-\frac{1}{k}e^{-k0}\right] \\
&= \lim_{b\to\infty}\left[-\frac{1}{k}e^{-kb}\right] + \frac{1}{k} \\
&= \frac{1}{k}
\end{aligned}$$

ii. Apply the Gram-Schmidt algorithm on $\mathcal{B}$ (in the given order) to find an orthogonal basis $\mathcal{B} = (h_1, h_2, h_3)$ for W.

Solution: We apply the Gram-Schmidt algorithm to obtain three new orthogonal functions h_1, h_2, h_3

- We start by setting $h_1 = e^{-x}$. At this point, we also evaluate:

$$\langle h_1, h_1 \rangle = \int_0^\infty e^{-x}e^{-x}dx = \int_0^\infty e^{-2x}dx = \frac{1}{2}$$

- To get h_2, we need to evaluate

$$h_2 = e^{-2x} - \frac{\langle e^{-2x}, h_1 \rangle}{\langle h_1, h_1 \rangle}h_1$$

From this, we have:

$$\langle e^{-2x}, h_1 \rangle = \int_0^\infty e^{-2x}e^{-x}dx = \int_0^\infty e^{-3x}dx = \frac{1}{3}$$

Hence,

$$h_2 = e^{-2x} - \frac{\frac{1}{3}}{\frac{1}{2}}e^{-x} = e^{-2x} - \frac{2}{3}e^{-x}$$

We also evaluate the norm:

$$\begin{aligned}\langle h_2, h_2\rangle &= \int_0^\infty \left(e^{-2x} - \frac{2}{3}e^{-x}\right)^2 dx \\ &= \int_0^\infty e^{-4x} - \frac{4}{3}e^{-3x} + \frac{4}{9}e^{-2x}dx \\ &= \frac{1}{4} - \frac{4}{3}\cdot\frac{1}{3} + \frac{4}{9}\cdot\frac{1}{2} = \frac{1}{36}\end{aligned}$$

- To get h_3, we need to evaluate

$$h_2 = e^{-4x} - \frac{\langle e^{-4x}, h_1\rangle}{\langle h_1, h_1\rangle}h_1 - \frac{\langle e^{-4x}, h_2\rangle}{\langle h_2, h_2\rangle}h_2$$

From this, we have:

$$\begin{aligned}\langle e^{-4x}, h_1\rangle &= \int_0^\infty e^{-4x}e^{-x}dx = \int_0^\infty e^{-5x}dx = \frac{1}{5} \\ \langle e^{-4x}, h_2\rangle &= \int_0^\infty e^{-4x}\left(e^{-2x} - \frac{2}{3}e^{-x}\right)dx \\ &= \int_0^\infty e^{-6x} - \frac{2}{3}e^{-5x}dx = \frac{1}{6} - \frac{2}{15} = \frac{1}{30}\end{aligned}$$

Hence,

$$\begin{aligned}h_3 &= e^{-4x} - \frac{\frac{1}{5}}{\frac{1}{2}}e^{-x} - \frac{\frac{1}{30}}{\frac{1}{36}}\left(e^{-2x} - \frac{2}{3}e^{-x}\right) \\ &= e^{-4x} - \frac{2}{5}e^{-x} - \frac{6}{5}\left(e^{-2x} - \frac{2}{3}e^{-x}\right) \\ &= e^{-4x} + \frac{2}{5}e^{-x} - \frac{6}{5}e^{-2x}\end{aligned}$$

So our three orthogonal functions are:

$$\begin{aligned}h_1 &= e^{-x} \\ h_2 &= e^{-2x} - \frac{2}{3}e^{-x} \\ h_3 &= e^{-4x} + \frac{2}{5}e^{-x} - \frac{6}{5}e^{-2x}\end{aligned}$$

Note: One can validate the solution to the differential equation by checking that $x^3 + 7x^2 + 14x + 8 = (x+1)(x+2)(x+4)$.

10 marks

2. In a simplified version of a popular online game, a team of Champions move between locations to protect their own Nexus, whilst attempting to destroy the opponent team's Nexus. We know the following about the locations and rules for movement:

   - A team consists of 5 Champions
   - There are five locations: *Home (Nexus)*,*Enemy (Nexus)*, *Safe lane*, *Mid lane*, and *Off lane*.
   - *Home* is adjacent to the *Safe lane*, *Mid lane*, and *Off lane*
   - *Enemy* is adjacent to the *Safe lane*, *Mid lane*, and *Off lane*
   - No other connections exist
   - All connections are bi-directional (Champions can move in both directions)
   - For safety reasons, Champions must move to an adjacent location at each time step and cannot stay in the same location
   - Champions move to adjacent locations with equal probabilities at each time step and cannot stay in the same location.

   If you wish, you can simplify the location using their respective starting letters. I.e. H,E,S,M,O.

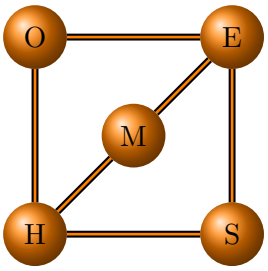
   Stylistic note: If you have a strong preference, you may use Ancient/Hero in place of Nexus/Champion.

   (a) Construct the Markov diagram for this game and show that the associate transition matrix T can be written as the following.

$$T = \begin{pmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \end{pmatrix}$$

   Hint: Be sure to state your ordering of the states and justify the entries in T

   Solution: The Markov diagram can be drawn as:



   We can use the given ordering: H,E,S,M,O as $s_1, \ldots, s_5$ respectively. Since H,E are connected to three different areas, their transition probabilities should be $\frac{1}{3}$. The other three areas (S,M,O) are connected to two areas and so their transition probabilities should be $\frac{1}{2}$.

   (b) Show that $\lambda = 1$ is an eigenvalue of T

Solution: The easiest way to validate the the result is to show $\det(T - I) = 0$. We start by row reducing T_I until we get to upper triangular form to simplify calculations:

$$T - I = \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & -1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & -1 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & -1 \end{pmatrix}$$

Applying row reduction gives:

$$\begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & -1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & -1 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{3} & -\frac{5}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{1}{6} & -\frac{5}{6} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} & -\frac{5}{6} \end{pmatrix} \begin{matrix} \\ \\ R_3 \to R_3 + \frac{1}{3}R_1 \\ R_4 \to R_4 + \frac{1}{3}R_1 \\ R_5 \to R_5 + \frac{1}{3}R_1 \end{matrix}$$

$$\sim \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \end{pmatrix} \begin{matrix} \\ \\ R_3 \to R_3 + \frac{1}{3}R_2 \\ R_4 \to R_4 + \frac{1}{3}R_2 \\ R_5 \to R_5 + \frac{1}{3}R_2 \end{matrix}$$

$$\sim \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{matrix} \\ \\ \\ R_4 \to R_4 + \frac{1}{2}R_3 \\ R_5 \to R_5 + \frac{1}{2}R_3 \end{matrix}$$

$$\sim \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad R_5 \to R_5 + R_4$$

By Proposition 4.16 and Theorem 2.35, we can say that $\det(T - I) = 0$ and hence $\lambda = 1$ is an eigenvalue.

Note: The characteristic polynomial $\det(T - \lambda I)$ is

$$\chi_T(\lambda) = x^5 - x^3$$

which has a root at $\lambda = 1$ (Thm 6.15)

(c) Determine an eigenvector associated with $\lambda = 1$ and hence find the stable distribution

Solution: We can use the REF from the previous part to find an eigenvector:

$$T - I \sim \begin{pmatrix} -1 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Using back substitution.

- s_5 is free, so let $s_5 = \alpha$
- $s_4 = s_5$, so $s_4 = \alpha$
- $\frac{2}{3}s_3 = \frac{1}{3}\alpha + \frac{1}{3}\alpha$. Solving, we get $s_3 = \alpha$
- $s_2 = \frac{1}{2}\alpha + \frac{1}{2}\alpha + \frac{1}{2}\alpha$. Solving, we get $s_2 = \frac{3}{2}\alpha$
- $s_1 = \frac{1}{2}\alpha + \frac{1}{2}\alpha + \frac{1}{2}\alpha$. Solving, we get $s_1 = \frac{3}{2}\alpha$

So an eigenvector of T associated with $\lambda = 1$ is $\left(\frac{3}{2}\alpha, \frac{3}{2}\alpha, \alpha, \alpha, \alpha\right)^T$. If we start with 5 Champions, then the corresponding eigenvector occurs when $\alpha = \frac{5}{6}$ to give $\left(\frac{5}{4}, \frac{5}{4}, \frac{5}{6}, \frac{5}{6}, \frac{5}{6}\right)^T$

(d) Prove, or provide a counter example: Let $\vec{v} \in \mathbb{R}^5$ be an initial distribution of Champions. The sequence of vectors $T^k\vec{v}$ must converge to the stable distribution

Hint: For this last part, you may use a computer program/software to assist your calculations. Be sure to state which parts were done using a computer and provide an outline of what you did.

Solution: Not all initial vectors converge to the stable distribution. A counter example might be $\vec{v} = (5, 0, 0, 0, 0)$ (everyone starting in the Home location). On odd steps, the probability of H,E are all zero. On even steps, the probability of S,M,O are all zero.

Additional details: The states are partitioned into two sets (H,E) and (S,M,O) and one must transition between sets at each step. i.e. it's impossible to stay within the same set. So if the initial vector is not evenly distributed over the two sets, then it will never reach the stable distribution.

9 marks

3. Interpreting large datasets is a necessary yet difficult challenge in most of the data science disciplines. Hence, dimensionality-reduction techniques are often used in data pre-processing, where the result should preserve the most significant features of the original dataset with minimal information loss. In this question, we will illustrate a dimensionality-reduction technique to determine the "most popular" show on a streaming service based on three metrics: the average rating, the average number of recommendations, and the retention rate.

	Name	Rating	Recommendations	Retention
1	*The Butcher of Blaviken*	4.0	2.0	0.60
2	*Salary Swindle*	4.2	2.1	0.59
3	*Unusual Objects*	3.9	2.0	0.58
4	*Calamari Contest*	4.3	2.1	0.62
5	*The Sicilian Defense*	4.1	2.2	0.63

The covariance matrix Σ contains the variances of the variables along the main diagonal and the covariances between each pair of variables in the other matrix positions as follows:

$$\Sigma = \begin{pmatrix} 0.025 & 0.0075 & 0.00175 \\ 0.0075 & 0.0070 & 0.00135 \\ 0.00175 & 0.00135 & 0.00043 \end{pmatrix}$$

For this question, quote numerical answers to at least 3 decimal points or 3 significant figures, whichever is more precise.

(a) Given that $\vec{v}_1 = (12.241, 4.462, 1)^T$ is an eigenvector of Σ. Find its associated eigenvalue λ_1.

Solution: Direct computation gives:

$$\Sigma\vec{v}_1 = \begin{pmatrix} 0.025 & 0.0075 & 0.00175 \\ 0.0075 & 0.0070 & 0.00135 \\ 0.00175 & 0.00135 & 0.00043 \end{pmatrix} \begin{pmatrix} 12.241 \\ 4.462 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.341 \\ 0.124 \\ 0.0279 \end{pmatrix} = 0.0279\,\vec{v}_1$$

Hence, the eigenvalue is 0.0279.

(b) The eigenvalues of Σ and their associated eigenvectors are summarised as follows:

Eigenvalues	Eigenvectors
λ_1: From Part (a)	$\vec{v}_1 = (12.241, 4.462, 1)^T$
$\lambda_2 = 0.000159$	$\vec{v}_2 = (-0.016, -0.180, 1)^T$
$\lambda_3 = 0.00439$	$\vec{v}_3 = (-2.184, 5.767, 1)^T$

Let $w_i = \dfrac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$ be the weight of each eigenvalue λ_i. Determine w_1 and show that λ_1 has the largest weight.

Solution: Direct computation gives: $w_1 = \dfrac{0.0279}{0.0279 + 0.000159 + 0.00439} = 0.979.$

Note that $w_1 + w_2 + w_3 = 1$ and each w_i is positive, then both w_2 and w_3 must be less than 0.979.

(c) We perform dimension reduction by projecting each data point onto the eigenspace $E_{\lambda_1}(\Sigma)$ of the largest weight eigenvalue λ_1.

Write each data point as a three dimensional vector $\vec{r}_i$ = (Rating, Recommendation, Retention), e.g, $\vec{r}_1 = (4.0, 2.0, 0.60)^T$. Compute $\pi_{\vec{v}_1}(\vec{r}_i)$, the orthogonal projection (w.r.t standard inner product) of each $\vec{r}_i$ onto the eigenspace of λ_1.

Solution: Using Theorem 5.27 to compute:

$$\begin{aligned}\pi_{\vec{v}_1}(\vec{r}_1) &= \frac{\langle \vec{r}_1, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 \\ &= \frac{\langle (4.0, 2.0, 0.6)^T, (12.241, 4.462, 1)^T \rangle}{\langle (12.241, 4.462, 1)^T, (12.241, 4.462, 1)^T \rangle} (12.241, 4.462, 1)^T \\ &= \frac{58.488}{170.752} (12.241, 4.462, 1)^T \\ &= 0.343(12.241, 4.462, 1)^T\end{aligned}$$

In a similar fashion, we calculate all $\pi_{\vec{w}_i}(\vec{v}_1)$:

- $\pi_{\vec{v}_1}(\vec{r}_1) = 0.343(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_2) = 0.359(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_3) = 0.335(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_4) = 0.367(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_5) = 0.355(12.241, 4.462, 1)^T$

(d) Interpret your result in part(c) to justify which show from the sample is "most popular".

Solution: Express each $\pi_{\vec{v}_1}(\vec{r}_i)$ as a multiple of $\vec{v}_1$ gives:

- $\pi_{\vec{v}_1}(\vec{r}_1) = 0.343(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_2) = 0.359(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_3) = 0.335(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_4) = 0.367(12.241, 4.462, 1)^T$
- $\pi_{\vec{v}_1}(\vec{r}_5) = 0.355(12.241, 4.462, 1)^T$

Since 0.367 is the highest coefficient among all, we conclude that scores highest along the direction of the eigenvecter. Hence, $\vec{r}_4$ (Calamari Contest) is the most popular show by these three metrics.

10 marks

4. Let V be an n-dimensional vector space. We aim to show that every linear map in the form of $T : V \to \mathbb{R}$ can be expressed in terms of an inner product $\langle , \rangle$ on V. More precisely:

$$\text{There exists some fixed } \vec{w} \in V,\ T(\vec{v}) = \langle \vec{v}, \vec{w} \rangle \text{ for every } \vec{v} \in V.$$

The proof is partitioned based on Im(T). But first, let's show that the mapping defined above is indeed a linear map.

(a) Let $\vec{w} \in V$. Show that $S : V \to \mathbb{R}$ such that $S(\vec{v}) = \langle \vec{v}, \vec{w} \rangle$ is a linear map.

Solution: S preserves addition
For every $\vec{v}_1, \vec{v}_2 \in V$:

$$\begin{aligned} S(\vec{v}_1 + \vec{v}_2) = \langle \vec{v}_1 + \vec{v}_2, \vec{w} \rangle &= \langle \vec{v}_1, \vec{w} \rangle + \langle \vec{v}_2, \vec{w} \rangle && \text{(by (iii) of Definition 5.11)} \\ &= S(\vec{v}_1) + S(\vec{v}_2) \end{aligned}$$

S preserves scalar multiplication
For every $\vec{v}_1 \in V$ and $c \in \mathbb{R}$:

$$\begin{aligned} S(c\vec{v}_1) = \langle c\vec{v}_1, \vec{w} \rangle &= c\langle \vec{v}_1, \vec{w} \rangle && \text{(by (iv) of Definition 5.11)} \\ &= cS(\vec{v}_1) \end{aligned}$$

(b) Let $T : V \to \mathbb{R}$ be a linear map. If Im(T) $= \{0\}$, show that $T(\vec{v}) = \langle \vec{v}, \vec{0} \rangle$ where $\vec{0} \in V$.

Solution: If Im(T) $= \{0\}$, then $T(\vec{v}) = 0$ for every $\vec{v} \in V$.
We claim that $\langle \vec{v}, \vec{0} \rangle = 0$ for every $\vec{v} \in V$. One way to show this is:

$$\begin{aligned} \langle \vec{v}, \vec{0} \rangle = \langle \vec{0}, \vec{v} \rangle &= \langle \vec{0} + \vec{0}, \vec{v} \rangle \\ &= \langle \vec{0}, \vec{v} \rangle + \langle \vec{0}, \vec{v} \rangle \\ &= 2\langle \vec{0}, \vec{v} \rangle \\ &= 2\langle \vec{v}, \vec{0} \rangle \end{aligned}$$

The only real number satisfying the condition $x = 2x$ is $x = 0$. so $\langle \vec{v}, \vec{0} \rangle = 0$.
Hence, $T(\vec{v}) = 0 = \langle \vec{v}, \vec{0} \rangle$ for every $\vec{v} \in V$

(c) Let $T : V \to \mathbb{R}$ be a linear map. If Im(T) $\neq \{0\}$:

i. Show that Ker(T) has dimension $n - 1$.
Hint: The only subspaces of $\mathbb{R}$ are $\{0\}$ and $\mathbb{R}$ itself.

Solution: Note that Im(T) must be a subspace of $\mathbb{R}$. If Im(T) $\neq \{0\}$, then Im(T) $= \mathbb{R}$. This implies that $\dim(\text{Im(T)}) = 1$. By Rank-Nullity Theorem, we have $\dim(\text{Ker(T)}) = \dim V - \dim(\text{Im(T)}) = n - 1$.

ii. Let $\mathcal{B} = (\vec{v}_1, \vec{v}_2, ..., \vec{v}_{n-1})$ be an orthonormal basis of Ker(T). We can extend $\mathcal{B}$ to an orthonormal basis of V by selecting $\vec{v}_n$ appropriately. That is, $\mathcal{C} = \mathcal{B} \cup \{\vec{v}_n\} = (\vec{v}_1, \vec{v}_2, ..., \vec{v}_{n-1}, \vec{v}_n)$ is an orthonormal basis of V. Show that T can be expressed as $T(\vec{v}) = \langle \vec{v}, \vec{w} \rangle$ for some fixed $\vec{w} \in V$.

Solution: Let $\vec{v} \in V$. Using Theorem 5.25, we can express $\vec{v}$ as a linear combination of basis $\mathcal{B}_2$ as follows:

$$\begin{aligned}\vec{v} &= \frac{\langle \vec{v}, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle}\vec{v}_1 + \frac{\langle \vec{v}, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle}\vec{v}_2 + \cdots + \frac{\langle \vec{v}, \vec{v}_n \rangle}{\langle \vec{v}_n, \vec{v}_n \rangle}\vec{v}_n \\ &= \langle \vec{v}, \vec{v}_1 \rangle \vec{v}_1 + \langle \vec{v}, \vec{v}_2 \rangle \vec{v}_2 + \cdots + \langle \vec{v}, \vec{v}_n \rangle \vec{v}_n \qquad (\text{each } \langle \vec{v}_i, \vec{v}_i \rangle = 1)\end{aligned}$$

Hence,

$$\begin{aligned}T(\vec{v}) &= T(\langle \vec{v}, \vec{v}_1 \rangle \vec{v}_1 + \langle \vec{v}, \vec{v}_2 \rangle \vec{v}_2 + \cdots + \langle \vec{v}, \vec{v}_n \rangle \vec{v}_n) \\ &= \langle \vec{v}, \vec{v}_1 \rangle T(\vec{v}_1) + \langle \vec{v}, \vec{v}_2 \rangle T(\vec{v}_2) + \cdots + \langle \vec{v}, \vec{v}_n \rangle T(\vec{v}_n) \\ &= \langle \vec{v}, \vec{v}_1 \rangle (0) + \langle \vec{v}, \vec{v}_2 \rangle (0) + \cdots + \langle \vec{v}, \vec{v}_n \rangle T(\vec{v}_n) \\ &\qquad\qquad (\text{since } \vec{v}_1, \vec{v}_2, ..., \vec{v}_{n-1} \in \text{Ker(T)}) \\ &= \langle \vec{v}, \vec{v}_n \rangle T(\vec{v}_n) \\ &= \langle \vec{v}, T(\vec{v}_n)\vec{v}_n \rangle \qquad (\text{as } T(\vec{v}_n) \in \mathbb{R} \text{ and by (iv) of Definition 5.11})\end{aligned}$$

Hence, we expressed $T(\vec{v}) = \langle \vec{v}, \vec{w} \rangle$ with $\vec{w} = T(\vec{v}_n)\vec{v}_n$.
This concludes the Riesz Representation Theorem, stating that every linear map in the form of $T : V \to \mathbb{R}$ can be expressed in terms of an inner product $\langle , \rangle$ on V. Furthermore, it can be shown that the choice of the fixed $\vec{w}$ for each T is unique.

2 (bonus) 5. Write a short passage/poem/story summing up your experiences with F18CF this semester. It will be graded on creativity, the accurate use of mathematical analogies, and the ability to make the grader laugh (or cringe).